

Notes: Che (RES, 2009)

A Dynamic Model of Privatization with Endogenous Post-Privatization Performance

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1 Big picture

What the paper wants to explain. Privatization has mixed empirical “after” effects: some firms improve, others deteriorate (at least initially). Che (2009) argues that this heterogeneity can arise *endogenously* from two forces:

1. **Institutions improve over time.** Private-property-rights protection strengthens gradually, so the incentive benefits of private ownership rise over time.
2. **Buyers are financially constrained.** Even if private ownership becomes socially efficient, the buyer may not be able to compensate the government at the efficient time, distorting *timing* (premature or delayed privatization).

A single unifying idea. Let t_s be the *efficient switching date* from state to private ownership (the first time private ownership becomes better in total surplus). Then:

- **Delayed privatization** ($t^e > t_s$) occurs when the buyer cannot afford to buy *at* t_s but can afford later; it generates *immediate improvement* in post-privatization performance.
- **Premature privatization** ($t^e < t_s$) occurs when waiting is infeasible; it generates *immediate deterioration*.
- **Dynamic IC constraints can force underpricing** (a price below both willingness-to-pay and ability-to-pay) even if the government has all bargaining power.

2 Static environment at each date t

Time is continuous, indexed by $t \geq 0$, with discount rate $r > 0$.

2.1 Technology and effort

At each instant, the firm generates:

- **normal profit** $\chi \equiv x\pi$, with parameters $x > 0$ and $\pi > 0$;
- **extra profit** π with probability $a \in [0, 1]$.

A manager/buyer M chooses effort a (interpretable as restructuring/monitoring). Effort cost is

$$c(a) = \frac{c}{2}a^2, \quad c > \pi, \quad (1)$$

so interior effort obtains in the relevant range.

2.2 Ownership regimes and institutional friction

There are two ownership regimes.

(i) Government ownership (G). If the extra profit is realized, M can hide a fraction $l \in (0, 1)$ of *that extra profit* as private benefit. Hence M privately captures $l\pi$ when the extra profit occurs.

(ii) Private ownership (P). Under private ownership, M is legally entitled to profits. But the government can *expropriate* with probability $1 - \mu(t)$ *before* M can hide any profit. Thus $\mu(t) \in [0, 1]$ measures the degree of protection of private property rights at time t .

Institutional development. $\mu(t)$ evolves deterministically and monotonically:

$$\mu(0) = 0, \quad \dot{\mu}(t) > 0 \text{ for } \mu(t) < 1, \quad \lim_{t \rightarrow \infty} \mu(t) = 1. \quad (2)$$

2.3 Financial constraint and bargaining

If privatization occurs at time t , M must pay price $q(t)$ to the government, subject to

$$q(t) \leq B, \quad (3)$$

where B is the buyer's exogenous financial capacity (assumed *not* expropriable by the government).

At each instant t , the government makes a take-it-or-leave-it offer $q(t)$. If M accepts, the firm becomes privately owned from that point on.

3 Effort choice and instantaneous payoffs

Given ownership at time t , M chooses a to maximize his instantaneous payoff net of effort cost.

3.1 Effort under government ownership

Under G , M gets $l\pi$ when the extra profit occurs (probability a), so

$$\max_{a \in [0,1]} a(l\pi) - \frac{c}{2}a^2.$$

The interior solution is

$$a^G = \frac{l\pi}{c}, \quad (4)$$

and M 's instantaneous payoff is

$$u_M^G = \frac{l^2\pi^2}{2c}. \quad (5)$$

The government's instantaneous payoff is normal profit plus the non-hidden share of extra profit:

$$u_G^G = x\pi + a^G(1-l)\pi = x\pi + \frac{l(1-l)\pi^2}{c}. \quad (6)$$

3.2 Effort under private ownership

Under P , with probability $\mu(t)$ the government does *not* expropriate, so M keeps profits. Thus M solves

$$\max_{a \in [0,1]} \mu(t)(x\pi + a\pi) - \frac{c}{2}a^2,$$

with interior solution

$$a^P(t) = \frac{\mu(t)\pi}{c}. \quad (7)$$

Then

$$u_M^P(t) = \mu(t)x\pi + \frac{\mu(t)^2\pi^2}{2c}, \quad (8)$$

and the government's instantaneous payoff under private ownership (coming from expropriation, probability $1 - \mu(t)$) is

$$u_G^P(t) = (1 - \mu(t))(x\pi + a^P(t)\pi) = (1 - \mu(t))x\pi + \frac{\mu(t)(1 - \mu(t))\pi^2}{c}. \quad (9)$$

4 Reduced-form trade-off objects: $w(t)$ and $v(t)$

Buyer/manager's instantaneous gain from privatization.

$$w(t) \equiv u_M^P(t) - u_M^G = \mu(t)x\pi + \frac{\mu(t)^2\pi^2}{2c} - \frac{l^2\pi^2}{2c}. \quad (10)$$

Government's instantaneous opportunity cost of privatization.

$$v(t) \equiv u_G^G - u_G^P(t) = \left(x\pi + \frac{l(1-l)\pi^2}{c}\right) - \left((1-\mu(t))x\pi + \frac{\mu(t)(1-\mu(t))\pi^2}{c}\right). \quad (11)$$

Interpretation. $w(t)$ captures how much better M does under private ownership at date t ; $v(t)$ captures how much better the government does by *keeping* state ownership at date t .

Lemma 1: qualitative shape

The paper proves:

- $w(t)$ is strictly increasing in t (because $\mu(t)$ increases), with $w(0) < 0$ and $\lim_{t \rightarrow \infty} w(t) > 0$.
- $v(t)$ is quasi-convex (U-shaped) in t ; it may initially fall when small increases in μ mainly raise incentives while expropriation remains likely, and eventually rises when expropriation becomes unlikely and the government loses rents under private ownership.
- $w(t) - v(t)$ is increasing and crosses zero exactly once.

5 Efficiency benchmark: the socially optimal switch date t_s

Total instantaneous surplus under ownership $o \in \{G, P\}$ is $u_M^o + u_G^o$. Hence the instantaneous social gain from privatizing at time t is

$$\Delta S(t) = (u_M^P(t) + u_G^P(t)) - (u_M^G + u_G^G) = w(t) - v(t).$$

Because $w(t) - v(t)$ crosses zero once, define the efficient switch date t_s by

$$w(t_s) = v(t_s) \iff \Delta S(t_s) = 0. \quad (12)$$

A useful algebraic simplification is

$$w(t) - v(t) = \frac{(\mu(t) - l)(2 - \mu(t) - l)\pi^2}{2c}, \quad (13)$$

so, since $2 - \mu - l > 0$ on $[0, 1]$, the unique crossing satisfies

$$\mu(t_s) = l. \quad (14)$$

Interpretation. The efficient time is exactly when private property rights become strong enough (μ) to match the “implicit incentive” the manager already has under state ownership (the diversion parameter l).

6 Dynamic bargaining objects

6.1 Irreversibility (Lemma 2)

Once privatization occurs, it is never reversed in equilibrium. The logic is monotonicity: private ownership becomes weakly more attractive to M over time because $\mu(t)$ rises, so there is no future date when both parties would agree to reverse.

6.2 PV values: willingness to pay and reservation value

Define present values of the *flow* differences:

$$W(t) \equiv \int_t^\infty e^{-r(z-t)} w(z) dz \quad (\text{buyer willingness to pay, PV}), \quad (15)$$

$$V(t) \equiv \int_t^\infty e^{-r(z-t)} v(z) dz \quad (\text{government reservation value, PV}). \quad (16)$$

Define also the annuitized objects $rW(t)$ and $rV(t)$.

6.3 The “first surplus” date t_0

Let t_0 satisfy

$$W(t_0) = V(t_0).$$

Because $W(t) - V(t)$ is increasing, t_0 is the earliest date at which discounted privatization surplus is nonnegative. The paper shows $t_0 < t_s$: discounted gains turn positive before the instantaneous gain turns positive.

6.4 Feasible privatization at time t

Privatization at time t is feasible iff

$$t \geq t_0 \quad \text{and} \quad B \geq V(t). \quad (17)$$

7 Equilibrium constraints (Markov subgame perfection)

At time t , a candidate privatization price must satisfy:

- (i) **Budget:** $q(t) \leq B$.
- (ii) **Participation:** $W(t) \geq q(t) \geq V(t)$.
- (iii) **No profitable delay for M :** for all $t' > t$,

$$W(t) - q(t) \geq e^{-r(t'-t)} (W(t') - q(t')).$$

- (iv) **No profitable delay for government:** for all $t' > t$,

$$q(t) - V(t) \geq e^{-r(t'-t)} (q(t') - V(t')).$$

7.1 Optional technical note: price dynamics from binding IC

Along any region where M 's no-delay constraint binds locally, differentiating yields an ODE:

$$\frac{dq(t)}{dt} = rq(t) - w(t). \quad (18)$$

Similarly, if the government's no-delay constraint binds locally:

$$\frac{dq(t)}{dt} = rq(t) - v(t). \quad (19)$$

These ODEs clarify why rB appears in the timing rules when the price is pinned at $q(t) = B$.

8 Equilibrium characterization: timing

Let $t^{\min}$ be a minimizer of $V(t)$ (exists by quasi-convexity/boundedness).

8.1 Proposition 1: no privatization

If the buyer's financial capacity is too small to cover the government's reservation value even at the cheapest feasible time, privatization never occurs. A practical sufficient condition is:

$$B < V(\max\{t_0, t^{\min}\}).$$

8.2 Proposition 2: premature privatization at the last feasible date

When B is high enough for some feasibility but not enough to remain feasible forever, there is a last feasible date t_1 such that

$$V(t_1) = B, \quad \left. \frac{dV}{dt} \right|_{t=t_1} > 0. \quad (20)$$

If $t_0 \leq t_1 < t_s$, then equilibrium privatization occurs at $t^e = t_1$ and the price is $q(t^e) = B$. This is "premature" because the efficient date t_s lies after feasibility ends.

8.3 Proposition 3: delayed privatization when $v(t_s) > rB$

If $t_s < t_1$ (so privatizing at t_s is feasible in PV terms), but the *instantaneous* opportunity cost is too high relative to the annuitized budget,

$$v(t_s) > rB,$$

then privatization is delayed to $t^e \in (t_s, t_1)$ where

$$v(t^e) = rB, \quad \left. \frac{dv}{dt} \right|_{t=t^e} < 0, \quad (21)$$

with price $q(t^e) = B$. Intuition: with $q = B$ fixed, delaying reduces the present value of the transfer at rate rB , so the government waits until $v(t)$ falls to that rate.

8.4 Propositions 4–5: efficient vs inefficient timing

Timing is inefficient iff

$$rB < \max\{v(t_s), rV(t_s)\}.$$

- **Delay** happens when $v(t_s) > \max\{rB, rV(t_s)\}$.
- **Premature privatization** happens when $rV(t_s) > \max\{rB, v(t_s)\}$.

Timing is efficient ($t^e = t_s$) iff

$$rB \geq \max\{v(t_s), rV(t_s)\}.$$

8.5 Corollary 1: budget regions

The paper splits cases depending on whether $v(t_s)$ is above or below $rV(t_s)$ (equivalently whether t_s lies before/after the minimum of v). In either case, the qualitative partition is:

- small B : no privatization;
- intermediate B : delayed or premature privatization;
- large B : efficient privatization at t_s .

9 Endogenous post-privatization performance

Produced expected profit at time t under ownership o is

$$\Pi^o(t) = x\pi + a^o(t)\pi.$$

Hence the immediate performance jump at equilibrium privatization time t^e is

$$\Delta\Pi^e \equiv \Pi^P(t^e) - \Pi^G(t^e) = (\mu(t^e) - l) \frac{\pi^2}{c}. \quad (22)$$

Proposition 6 (sign of the performance change).

- If $t^e > t_s$ (delay), then $\mu(t^e) > l$ and $\Delta\Pi^e > 0$.
- If $t^e < t_s$ (premature), then $\mu(t^e) < l$ and $\Delta\Pi^e < 0$.
- If $t^e = t_s$ (efficient), then $\Delta\Pi^e = 0$ (marginal at the switch).

Corollary 2. Conditional on privatization, $|\Delta\Pi^e|$ decreases with B because the equilibrium timing distortion shrinks as the buyer becomes less constrained.

10 Comparative statics: which firms improve vs deteriorate?

The paper maps primitives into regions via the threshold curves that appear in Figures 4–6.

10.1 Proposition 7: heterogeneity in x (importance of normal profits)

When x is larger, normal profits are larger relative to effort-sensitive profits, so effort matters less. Under a regularity condition (existence of x^* with $v(t_s, x^*) = rV(t_s, x^*)$), the paper shows:

- larger x makes outcomes weakly less efficient (inefficiency more likely);
- firms that deteriorate after privatization are those where effort is less important (larger x) relative to those that improve;

- in the deterioration (premature) region, larger x implies earlier privatization and a larger decline;
- in the improvement (delay) region, larger x implies longer delay and a larger improvement.

10.2 Proposition 8: heterogeneity in firm size π

Assuming existence of π^* with $v(t_s, \pi^*) = rV(t_s, \pi^*)$, the paper shows:

- larger firms (higher π) are more likely to be privatized with improved performance;
- smaller firms are more likely to be privatized with deteriorating performance (if privatized at all).

10.3 Proposition 9: heterogeneity in managerial productivity c

Because $v(t_s, c)$ is constant in c while $rV(t_s, c)$ rises with c , there exists c^* with $v(t_s, c^*) = rV(t_s, c^*)$. Then:

- if a firm with cost c is privatized, then any firm with *more productive* management (lower c) is also privatized;
- firms with low c are associated with delayed privatization and improvement;
- firms with high c are associated with premature privatization and deterioration.

11 Pricing and underpricing

11.1 Proposition 10: equilibrium price

(a) **Inefficient timing.** If timing is inefficient (delayed or premature), the equilibrium price exhausts the buyer's capacity:

$$q(t^e) = B.$$

(b) **Efficient timing.** If timing is efficient ($t^e = t_s$), the equilibrium price is unique and typically satisfies

$$q(t_s) < \min\{B, W(t_s)\},$$

except:

- if the buyer's budget is slack for all t ($B \geq W(t)$), then the government can charge $q(t_s) = W(t_s)$;
- in a knife-edge case where $rB = w(t_s)$, efficient timing can occur with $q(t_s) = B$.

Why underpricing can happen even with full government bargaining power. Because $\mu(t)$ increases, $W(t)$ tends to increase, so the buyer has an incentive to wait. To implement the efficient timing, the government must offer the buyer enough rent today so that delaying is not profitable; this creates underpricing relative to the static surplus.

11.2 Corollary 3: who captures surplus?

The government captures a positive share of the total gains from trade in any equilibrium *except* when privatization occurs at the last feasible date t_1 (the “premature-at- t_1 ” case), in which case $q(t_1) = B = V(t_1)$ and the buyer captures the surplus.

12 Institutional speed

12.1 Proposition 11: faster institutional development can worsen outcomes

Under a parametric speed version (e.g. $\mu(t) = 1 - e^{-\eta t}$), faster institutional development (higher η) can:

- turn privatization from feasible to infeasible (reservation value grows too quickly relative to B), or
- turn efficient privatization into premature privatization (to avoid losing feasibility).

13 Figures

Important note. This is a theoretical paper with **no empirical tables**. All visual evidence is via figures.

Figure 1: $w(t)$ and $v(t)$

Illustrates Lemma 1 and the efficient switch date t_s . $w(t)$ increases (buyer increasingly prefers private ownership); $v(t)$ is U-shaped and positive (government prefers state ownership pointwise). They intersect once at t_s .

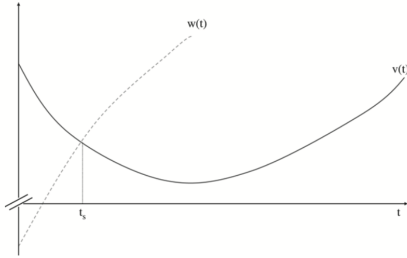


FIGURE 1
Instantaneous pay-off differences between government ownership and private ownership

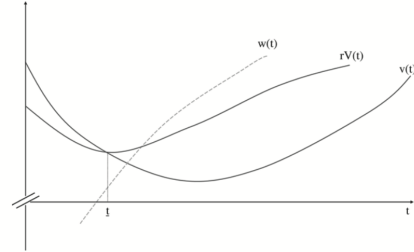


FIGURE 2
The annualized cost of privatization to G vs. the instantaneous cost of privatization to G

Figure 2: $W(t)$, $V(t)$, and feasibility under a budget B

Shows $W(t)$ increasing and $V(t)$ quasi-convex. t_0 is where W crosses V (discounted gains become positive). A horizontal B line yields feasible dates where $V(t) \leq B$, and the last intersection on the rising side is t_1 .

Figure 3: delayed privatization

Plots $w(t)$, $v(t)$, $rV(t)$, and the horizontal rB line. Delay arises when $v(t_s) > rB$ and v is decreasing: equilibrium waits until $v(t) = rB$.

Figure 4: regions in (x, rB)

Vertical axis: rB ; horizontal axis: x . The boundary curves are:

- $rV(\max\{t_0, t^{\min}\}, x)$ separating “no privatization” from “privatization”,
- $v(t_s, x)$ and $rV(t_s, x)$ separating efficient from delayed/premature regions.

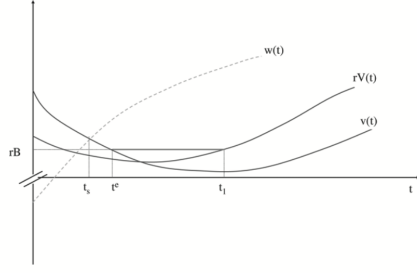


FIGURE 3
Delayed privatization

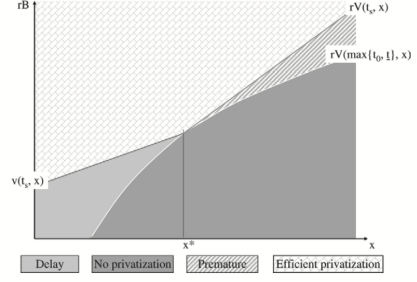


FIGURE 4
Privatization outcomes in relation to x, rB

Figure 5: regions in (π, rB)

Two panels reflect that the relative ordering of $v(t_s, \pi)$ and $rV(t_s, \pi)$ can differ by parameterization. The key takeaway: higher π shifts the economy toward the delayed/improvement region.

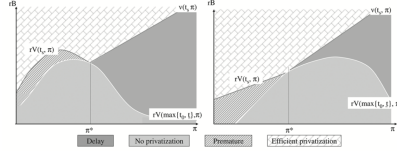


FIGURE 5
Privatization outcomes in relation to x, rB

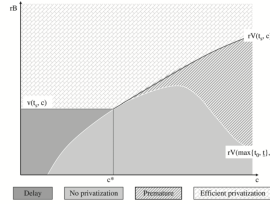


FIGURE 6
Privatization outcomes in relation to c, rB

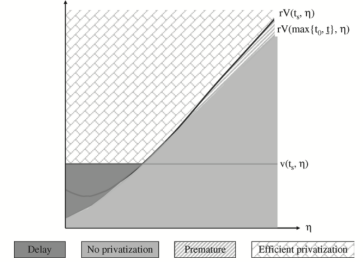


FIGURE A1
Faster institutional development can increase the annualized costs of privatization

Figure 6: regions in (c, rB)

Because $v(t_s, c)$ is flat in c while $rV(t_s, c)$ increases in c , higher c pushes outcomes toward premature privatization and deterioration.

Figure A1: regions in (η, rB)

As η rises, the annuitized cost curves can shift upward, shrinking the efficient region and expanding premature/no-privatization regions (Proposition 11).

14 Some equations

- Efficient switch time: $\mu(t_s) = l$.
- Performance jump: $\Delta\Pi^e = (\mu(t^e) - l)\pi^2/c$.
- Inefficiency condition: $rB < \max\{v(t_s), rV(t_s)\}$.
- Price: $q = B$ if timing inefficient; $q < \min\{B, W(t_s)\}$ if timing efficient (generically).

15 Conclusion

15.1 Summary: what the paper concludes

Che (2009) shows that the mixed empirical evidence on post-privatization performance can arise *endogenously* from a dynamic interaction between (i) **gradual institutional development** (private property-rights protection improves over time) and (ii) **buyer financial constraints** (the manager/buyer cannot always pay the government’s reservation value at the socially optimal time). In the model, the socially efficient privatization time t_s is pinned down by an institutional threshold:

$$\mu(t_s) = l,$$

i.e., privatization becomes efficient exactly when the quality of private property-rights protection $\mu(t)$ reaches the “implicit incentive” already embedded in state ownership via diversion/private benefits l .

However, equilibrium privatization need not occur at t_s . Because the buyer faces a hard budget limit B and because both parties can delay, equilibrium timing can be **distorted**:

- If B is too small, privatization may **never occur**.
- If B is intermediate, privatization can be **premature** (forced to occur before t_s to avoid losing feasibility) or **delayed** (waiting until the government’s effective flow cost is low enough relative to the annuitized budget).
- If B is large enough, privatization occurs **efficiently** at t_s .

Thus, the paper’s main conclusion is that *the timing distortion is the key state variable that determines whether performance improves or deteriorates after privatization*, even when the government has all bargaining power at the moment of sale.

15.2 Core intuition: why post-privatization performance is endogenous

The mechanism is a single chain:

1. **Incentives differ across regimes.** Under state ownership, the manager can divert a fraction l of the “extra” profit, which provides an incentive to exert effort. Under private ownership, the manager can keep profits only with probability $\mu(t)$ because of expropriation risk, so effort incentives depend on $\mu(t)$.
2. **Institutions improve over time** ($\mu(t) \uparrow$). This makes private ownership increasingly attractive to the buyer over time, and it also changes the government’s reservation value.
3. **The buyer cannot always pay at the efficient time.** Even if society would like to switch at t_s (when $\mu(t_s) = l$), the buyer may not be able to transfer enough surplus to satisfy the government. That financial wedge forces the equilibrium sale date t^e away from t_s .
4. **The sign of the performance change is pinned down by timing.** The paper delivers an explicit expression for the immediate performance change at privatization:

$$\Delta\Pi^e \equiv \Pi^P(t^e) - \Pi^G(t^e) = (\mu(t^e) - l) \frac{\pi^2}{c}.$$

Therefore:

- If privatization is **delayed** ($t^e > t_s$), then $\mu(t^e) > l$ and $\Delta\Pi^e > 0$ (performance improves).
- If privatization is **premature** ($t^e < t_s$), then $\mu(t^e) < l$ and $\Delta\Pi^e < 0$ (performance deteriorates).
- If privatization is **efficient** ($t^e = t_s$), then $\Delta\Pi^e = 0$ at the margin.

15.3 Pricing conclusion: why “underpricing” can happen even with full government bargaining power

A second major conclusion is that privatization can be **underpriced** even if the government makes take-it-or-leave-it offers. When the efficient timing t_s is implementable, the government may still need to set a price below static willingness-to-pay to prevent the buyer from waiting, because the buyer’s outside option improves as institutions improve ($W(t)$ rises with t). In other words, *dynamic incentive compatibility* can force the seller to leave rents to the buyer to avoid inefficient delay. By contrast, when timing is inefficient (premature or delayed), the equilibrium price typically hits the buyer’s constraint, $q(t^e) = B$.

15.4 Testable predictions and how to use them in empirical work

The model generates sharp empirical predictions that link (i) *timing*, (ii) *financial constraints*, and (iii) *institutional quality* to outcomes:

- **Heterogeneous post-privatization effects:** some firms improve immediately, others deteriorate, and this heterogeneity is systematic.
- **Performance improvements are more likely when privatization occurs in “high- μ ” environments** (better property-rights protection at the time of sale).
- **Financial capacity matters:** the absolute performance jump $|\Delta\Pi^e|$ declines as B rises (distortions shrink when the buyer is less constrained).
- **Selection patterns:** larger firms and more productive managers are more likely to be privatized and more likely to exhibit improvements, while small firms/high-cost managers are more likely to be prematurely privatized (if privatized at all) and exhibit deterioration.

15.5 Takeaway

Privatization outcomes are not mechanically “good” or “bad”: they depend on whether financial constraints push the sale before or after the efficient institutional threshold, and that timing distortion endogenously determines the sign and magnitude of post-privatization performance changes.